



PREALG: Fractions and Decimals

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Learning Objectives

- Add and subtract fractions with unlike denominators
- Multiply and divide fractions and mixed numbers
- Convert between fractions, decimals, and percents
- Simplify fractions to lowest terms

Simplify each expression completely. Show all steps and circle your final answer.

Adding fractions

1. Add the fractions and simplify: $\frac{1}{2} + \frac{3}{3}$.

$$\frac{1}{2} + \frac{3}{3}$$

Answer: _____

2. A recipe calls for $\frac{1}{4}$ cup of flour and $\frac{3}{3}$ cup of sugar. How much total dry ingredient is needed?

$$\frac{1}{4} + \frac{3}{3}$$

Answer: _____

3. Add: $\frac{4}{7} + \frac{5}{8}$. Simplify your answer completely.

$$\frac{4}{7} + \frac{5}{8}$$

Answer: _____

4. Add the fractions and simplify: $\frac{2}{5} + \frac{5}{4}$.

$$\frac{2}{5} + \frac{5}{4}$$

Answer: _____

5. A recipe calls for $\frac{1}{3}$ cup of flour and $\frac{1}{4}$ cup of sugar. How much total dry ingredient is needed?

$$\frac{1}{3} + \frac{1}{4}$$

Answer: _____



6. Add: $\frac{5}{3} + \frac{4}{8}$. Simplify your answer completely.

$$\frac{5}{3} + \frac{4}{8}$$

Answer: _____

7. Add the fractions and simplify: $\frac{1}{4} + \frac{4}{3}$.

$$\frac{1}{4} + \frac{4}{3}$$

Answer: _____

8. A recipe calls for $\frac{1}{2}$ cup of flour and $\frac{2}{5}$ cup of sugar. How much total dry ingredient is needed?

$$\frac{1}{2} + \frac{2}{5}$$

Answer: _____

9. Add: $\frac{4}{8} + \frac{5}{3}$. Simplify your answer completely.

$$\frac{4}{8} + \frac{5}{3}$$

Answer: _____

Dividing fractions

10. Divide the fractions: $\frac{3}{5} \div \frac{4}{4}$. Use Keep-Change-Flip.

$$\frac{3}{5} \div \frac{4}{4}$$

Answer: _____

11. Divide: $\frac{1}{5} \div \frac{1}{6}$. (Keep the first fraction, Change division to multiplication, Flip the second fraction.)

$$\frac{1}{5} \div \frac{1}{6}$$

Answer: _____

12. You have $\frac{4}{3}$ of a pizza. If each serving is $\frac{2}{8}$ of the whole pizza, how many servings do you have?

$$\frac{4}{3} \div \frac{2}{8}$$

Answer: _____



13. Divide the fractions: $\frac{1}{4} \div \frac{5}{6}$. Use Keep-Change-Flip.

$$\frac{1}{4} \div \frac{5}{6}$$

Answer: _____

14. Divide: $\frac{5}{3} \div \frac{2}{7}$. (Keep the first fraction, Change division to multiplication, Flip the second fraction.)

$$\frac{5}{3} \div \frac{2}{7}$$

Answer: _____

15. You have $\frac{3}{3}$ of a pizza. If each serving is $\frac{3}{7}$ of the whole pizza, how many servings do you have?

$$\frac{3}{3} \div \frac{3}{7}$$

Answer: _____

16. Divide the fractions: $\frac{6}{2} \div \frac{1}{8}$. Use Keep-Change-Flip.

$$\frac{6}{2} \div \frac{1}{8}$$

Answer: _____

17. Divide: $\frac{1}{4} \div \frac{6}{5}$. (Keep the first fraction, Change division to multiplication, Flip the second fraction.)

$$\frac{1}{4} \div \frac{6}{5}$$

Answer: _____

18. You have $\frac{4}{3}$ of a pizza. If each serving is $\frac{2}{5}$ of the whole pizza, how many servings do you have?

$$\frac{4}{3} \div \frac{2}{5}$$

Answer: _____

Multiplying fractions

19. Multiply the fractions and simplify: $\frac{5}{4} \times \frac{2}{6}$.

$$\frac{5}{4} \times \frac{2}{6}$$

Answer: _____



20. Multiply: $\frac{1}{4} \times \frac{4}{3}$. Simplify your answer.

$$\frac{1}{4} \times \frac{4}{3}$$

Answer: _____

21. A recipe calls for $\frac{2}{3}$ cup of sugar. If you are making $\frac{2}{4}$ of the recipe, how many cups of sugar do you need?

$$\frac{2}{3} \times \frac{2}{4}$$

Answer: _____

22. Multiply the fractions and simplify: $\frac{3}{4} \times \frac{5}{3}$.

$$\frac{3}{4} \times \frac{5}{3}$$

Answer: _____

23. Multiply: $\frac{4}{3} \times \frac{5}{4}$. Simplify your answer.

$$\frac{4}{3} \times \frac{5}{4}$$

Answer: _____

24. A recipe calls for $\frac{3}{2}$ cup of sugar. If you are making $\frac{1}{2}$ of the recipe, how many cups of sugar do you need?

$$\frac{3}{2} \times \frac{1}{2}$$

Answer: _____

Simplifying fractions

25. Simplify the fraction $\frac{6}{12}$ to lowest terms.

$$\frac{6}{12}$$

Answer: _____

26. A class has 16 students present out of 12 enrolled. Write this fraction in lowest terms.

$$\frac{16}{12}$$

Answer: _____



27. Simplify $35/15$. Is the result a proper fraction, an improper fraction, or a whole number?

$$\frac{35}{15}$$

Answer: _____

28. Simplify the fraction $16/36$ to lowest terms.

$$\frac{16}{36}$$

Answer: _____

29. A class has 20 students present out of 12 enrolled. Write this fraction in lowest terms.

$$\frac{20}{12}$$

Answer: _____

30. Simplify $24/16$. Is the result a proper fraction, an improper fraction, or a whole number?

$$\frac{24}{16}$$

Answer: _____





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ANSWER KEY & SOLUTIONS

Topics: Adding fractions, Multiplying fractions, Simplifying fractions, Dividing fractions. All answers verified by independent computation.

Solutions



Adding fractions

1. Add the fractions and simplify: $\frac{1}{2} + \frac{3}{3}$.

$$\frac{1}{2} + \frac{3}{3}$$

→ Find a common denominator, then add numerators. Simplify if possible.

→ Result: $\frac{3}{2}$.

Answer: $= \frac{3}{2}$

2. A recipe calls for $\frac{1}{4}$ cup of flour and $\frac{3}{3}$ cup of sugar. How much total dry ingredient is needed?

$$\frac{1}{4} + \frac{3}{3}$$

→ Total = $\frac{1}{4} + \frac{3}{3} = \frac{5}{4}$ cups.

Answer: $= \frac{5}{4}$

3. Add: $\frac{4}{7} + \frac{5}{8}$. Simplify your answer completely.

$$\frac{4}{7} + \frac{5}{8}$$

→ Find the LCD of 7 and 8: LCD = 56.

→ Rewrite each fraction with denominator 56.

→ Add numerators and simplify: $\frac{67}{56}$.

Answer: $= \frac{67}{56}$

4. Add the fractions and simplify: $\frac{2}{5} + \frac{5}{4}$.

$$\frac{2}{5} + \frac{5}{4}$$

→ Find a common denominator, then add numerators. Simplify if possible.

→ Result: $\frac{33}{20}$.

Answer: $= \frac{33}{20}$

5. A recipe calls for $\frac{1}{3}$ cup of flour and $\frac{1}{4}$ cup of sugar. How much total dry ingredient is needed?

$$\frac{1}{3} + \frac{1}{4}$$

→ Total = $\frac{1}{3} + \frac{1}{4} = \frac{7}{12}$ cups.

Answer: $= \frac{7}{12}$

6. Add: $\frac{5}{3} + \frac{4}{8}$. Simplify your answer completely.

$$\frac{5}{3} + \frac{4}{8}$$

→ Find the LCD of 3 and 8: LCD = 24.

→ Rewrite each fraction with denominator 24.

→ Add numerators and simplify: $\frac{13}{6}$.

Answer: $= \frac{13}{6}$



7. Add the fractions and simplify: $\frac{1}{4} + \frac{4}{3}$.

$$\frac{1}{4} + \frac{4}{3}$$

→ Find a common denominator, then add numerators. Simplify if possible.

→ Result: $\frac{19}{12}$.

Answer: $= \frac{19}{12}$

8. A recipe calls for $\frac{1}{2}$ cup of flour and $\frac{2}{5}$ cup of sugar. How much total dry ingredient is needed?

$$\frac{1}{2} + \frac{2}{5}$$

→ Total = $\frac{1}{2} + \frac{2}{5} = \frac{9}{10}$ cups.

Answer: $= \frac{9}{10}$

9. Add: $\frac{4}{8} + \frac{5}{3}$. Simplify your answer completely.

$$\frac{4}{8} + \frac{5}{3}$$

→ Find the LCD of 8 and 3: LCD = 24.

→ Rewrite each fraction with denominator 24.

→ Add numerators and simplify: $\frac{13}{6}$.

Answer: $= \frac{13}{6}$



Dividing fractions

10. Divide the fractions: $\frac{3}{5} \div \frac{4}{4}$. Use Keep-Change-Flip.

$$\frac{3}{5} \div \frac{4}{4}$$

→ Keep $\frac{3}{5}$, Change div to x, Flip $\frac{4}{4}$ to $\frac{4}{4}$.

→ = $\frac{3}{5} \times \frac{4}{4}$. Simplify: $\frac{3}{5}$.

Answer: = $\frac{3}{5}$

11. Divide: $\frac{1}{5} \div \frac{1}{6}$. (Keep the first fraction, Change division to multiplication, Flip the second fraction.)

$$\frac{1}{5} \div \frac{1}{6}$$

→ Keep-Change-Flip: $\frac{1}{5} \times \frac{6}{1}$.

→ Multiply and simplify: $\frac{6}{5}$.

Answer: = $\frac{6}{5}$

12. You have $\frac{4}{3}$ of a pizza. If each serving is $\frac{2}{8}$ of the whole pizza, how many servings do you have?

$$\frac{4}{3} \div \frac{2}{8}$$

→ Divide: $\frac{4}{3} \div \frac{2}{8}$.

→ Keep-Change-Flip: $\frac{4}{3} \times \frac{8}{2}$.

→ Simplify: $\frac{16}{3}$ servings.

Answer: = $\frac{16}{3}$

13. Divide the fractions: $\frac{1}{4} \div \frac{5}{6}$. Use Keep-Change-Flip.

$$\frac{1}{4} \div \frac{5}{6}$$

→ Keep $\frac{1}{4}$, Change div to x, Flip $\frac{5}{6}$ to $\frac{6}{5}$.

→ = $\frac{1}{4} \times \frac{6}{5}$. Simplify: $\frac{3}{10}$.

Answer: = $\frac{3}{10}$

14. Divide: $\frac{5}{3} \div \frac{2}{7}$. (Keep the first fraction, Change division to multiplication, Flip the second fraction.)

$$\frac{5}{3} \div \frac{2}{7}$$

→ Keep-Change-Flip: $\frac{5}{3} \times \frac{7}{2}$.

→ Multiply and simplify: $\frac{35}{6}$.

Answer: = $\frac{35}{6}$

15. You have $\frac{3}{3}$ of a pizza. If each serving is $\frac{3}{7}$ of the whole pizza, how many servings do you have?

$$\frac{3}{3} \div \frac{3}{7}$$

→ Divide: $\frac{3}{3} \div \frac{3}{7}$.

→ Keep-Change-Flip: $\frac{3}{3} \times \frac{7}{3}$.

→ Simplify: $\frac{7}{3}$ servings.

Answer: = $\frac{7}{3}$



16. Divide the fractions: $\frac{6}{2} \div \frac{1}{8}$. Use Keep-Change-Flip.

$$\frac{6}{2} \div \frac{1}{8}$$

→ Keep 6/2, Change div to x, Flip 1/8 to 8/1.

→ = $\frac{6}{2} \times \frac{8}{1}$. Simplify: 24.

Answer: = 24

17. Divide: $\frac{1}{4} \div \frac{6}{5}$. (Keep the first fraction, Change division to multiplication, Flip the second fraction.)

$$\frac{1}{4} \div \frac{6}{5}$$

→ Keep-Change-Flip: $\frac{1}{4} \times \frac{5}{6}$.

→ Multiply and simplify: $\frac{5}{24}$.

Answer: = $\frac{5}{24}$

18. You have $\frac{4}{3}$ of a pizza. If each serving is $\frac{2}{5}$ of the whole pizza, how many servings do you have?

$$\frac{4}{3} \div \frac{2}{5}$$

→ Divide: $\frac{4}{3} \div \frac{2}{5}$.

→ Keep-Change-Flip: $\frac{4}{3} \times \frac{5}{2}$.

→ Simplify: 10/3 servings.

Answer: = $\frac{10}{3}$



Multiplying fractions

19. Multiply the fractions and simplify: $\frac{5}{4} \times \frac{2}{6}$.

$$\frac{5}{4} \times \frac{2}{6}$$

→ Multiply numerators: $5 \times 2 = 10$.

→ Multiply denominators: $4 \times 6 = 24$.

→ Simplify $10/24$: $5/12$.

Answer: $\frac{10}{24} = 5/12$

20. Multiply: $\frac{1}{4} \times \frac{4}{3}$. Simplify your answer.

$$\frac{1}{4} \times \frac{4}{3}$$

→ Multiply numerators: $1 \times 4 = 4$.

→ Multiply denominators: $4 \times 3 = 12$.

→ Simplify $4/12$: $1/3$.

Answer: $\frac{4}{12} = 1/3$

21. A recipe calls for $\frac{2}{3}$ cup of sugar. If you are making $\frac{2}{4}$ of the recipe, how many cups of sugar do you need?

$$\frac{2}{3} \times \frac{2}{4}$$

→ Multiply the fractions: $\frac{2}{3} \times \frac{2}{4}$.

→ Numerator: $2 \times 2 = 4$. Denominator: $3 \times 4 = 12$.

→ Simplify: $\frac{1}{3}$ cups.

Answer: $\frac{4}{12} = 1/3$

22. Multiply the fractions and simplify: $\frac{3}{4} \times \frac{5}{3}$.

$$\frac{3}{4} \times \frac{5}{3}$$

→ Multiply numerators: $3 \times 5 = 15$.

→ Multiply denominators: $4 \times 3 = 12$.

→ Simplify $15/12$: $5/4$.

Answer: $\frac{15}{12} = 5/4$

23. Multiply: $\frac{4}{3} \times \frac{5}{4}$. Simplify your answer.

$$\frac{4}{3} \times \frac{5}{4}$$

→ Multiply numerators: $4 \times 5 = 20$.

→ Multiply denominators: $3 \times 4 = 12$.

→ Simplify $20/12$: $5/3$.

Answer: $\frac{20}{12} = 5/3$



24. A recipe calls for $\frac{3}{2}$ cup of sugar. If you are making $\frac{1}{2}$ of the recipe, how many cups of sugar do you need?

$$\frac{3}{2} \times \frac{1}{2}$$

→ Multiply the fractions: $\frac{3}{2} \times \frac{1}{2}$.

→ Numerator: $3 \times 1 = 3$. Denominator: $2 \times 2 = 4$.

→ Simplify: $\frac{3}{4}$ cups.

Answer: $\frac{3}{4} = \frac{3}{4}$



Simplifying fractions

25. Simplify the fraction 6/12 to lowest terms.

$$\frac{6}{12}$$

→ Find the GCF of 6 and 12. The GCF is 3.

→ Divide numerator and denominator by 3: $6 \div 3 = 2$ and $12 \div 3 = 4$.

→ Simplified fraction: 2/4.

Answer: $= \frac{6 \div 3}{12 \div 3} = 2/4$

26. A class has 16 students present out of 12 enrolled. Write this fraction in lowest terms.

$$\frac{16}{12}$$

→ Write the fraction: 16/12.

→ Find the GCF of 16 and 12: GCF = 4.

→ Divide both by 4: 4/3.

Answer: $= \frac{16 \div 4}{12 \div 4} = 4/3$

27. Simplify 35/15. Is the result a proper fraction, an improper fraction, or a whole number?

$$\frac{35}{15}$$

→ Find the GCF of 35 and 15: GCF = 5.

→ Divide: $35 \div 5 = 7$, $15 \div 5 = 3$.

→ Simplified: 7/3. Since the numerator is larger than the denominator, this is an improper fraction.

Answer: $= \frac{35 \div 5}{15 \div 5} = 7/3$

28. Simplify the fraction 16/36 to lowest terms.

$$\frac{16}{36}$$

→ Find the GCF of 16 and 36. The GCF is 4.

→ Divide numerator and denominator by 4: $16 \div 4 = 4$ and $36 \div 4 = 9$.

→ Simplified fraction: 4/9.

Answer: $= \frac{16 \div 4}{36 \div 4} = 4/9$

29. A class has 20 students present out of 12 enrolled. Write this fraction in lowest terms.

$$\frac{20}{12}$$

→ Write the fraction: 20/12.

→ Find the GCF of 20 and 12: GCF = 4.

→ Divide both by 4: 5/3.

Answer: $= \frac{20 \div 4}{12 \div 4} = 5/3$



30. Simplify $24/16$. Is the result a proper fraction, an improper fraction, or a whole number?

$$\frac{24}{16}$$

→ Find the GCF of 24 and 16: $GCF = 4$.

→ Divide: $24/4 = 6$, $16/4 = 4$.

→ Simplified: $6/4$. Since the numerator is larger than the denominator, this is an improper fraction.

Answer: $= \frac{24 \div 4}{16 \div 4} = 6/4$

